

FINAL PROJECT DEEP LEARNING: SKIN DISEASES CLASSIFICATION

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INTRODUCTION

As students of deep learning, we wanted our final project to tackle a problem with real-world impact. Skin cancer is one of the most significant cancers worldwide, and early diagnosis is a critical factor in improving patient survival rates. However, dermatological diagnosis from dermoscopic images is a challenging task even for experienced clinicians, given the high visual similarity between lesion types and the significant inter-observer variability reported in practice.

This project addresses the problem of automated multi-class skin lesion classification using the HAM10000 dataset, which contains 10,015 dermoscopic images spanning seven diagnostic categories: melanocytic nevi (*nv*), melanoma (*mel*), benign keratosis-like lesions (*bkl*), basal cell carcinoma (*bcc*), actinic keratoses (*akiec*), vascular lesions (*vasc*), and dermatofibroma (*df*). The dataset presents a challenging scenario due to its strong class imbalance: nevus accounts for the majority of samples, while clinically relevant malignant classes such as melanoma and BCC are underrepresented. Our goal is to build and evaluate deep learning models capable of discriminating between these seven classes, and providing model interpretability through gradient-based visualisation techniques.

STATE OF THE ART

Convolutional neural networks have become the standard approach for dermoscopic image analysis since Esteva et al. (2017) demonstrated dermatologist-level performance using a fine-tuned Inception network. Among the architectures explored on HAM10000, EfficientNet-based models have shown strong results due to their compound scaling strategy, which simultaneously optimises depth, width, and resolution, achieving a favourable accuracy-to-parameter ratio compared to ResNet and VGG families.

Explainable AI is one of the most critical components in a MLOps pipeline. Without it, it is difficult to assess what the model has actually learned and whether its predictions are based on meaningful patterns. This is particularly relevant in the medical sector, where a model's decision can directly impact patient outcomes. Unlike other domains, medicine demands not only accurate predictions but also interpretable ones: a clinician needs to understand why a model flagged a certain diagnosis before acting on it, as blindly trusting a black-box system could lead to misdiagnoses or overlooked conditions with serious consequences for the patient. To address this, we have integrated GradCAM (Gradient-weighted Class Activation Mapping), which highlights the specific image regions the model focuses on when classifying each disease category.

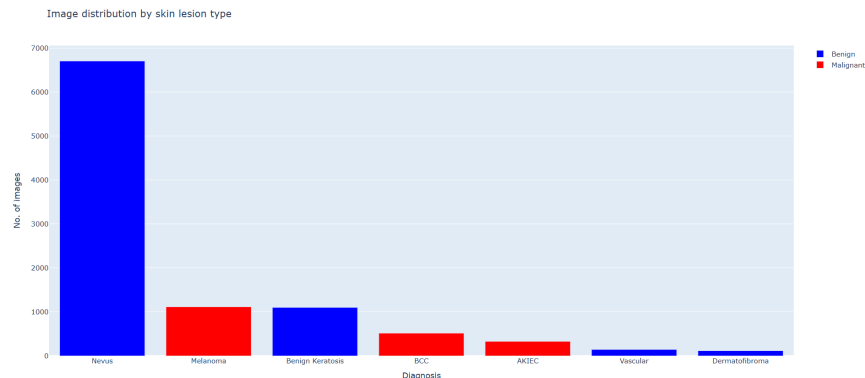
METHODOLOGY

1. Data Analysis

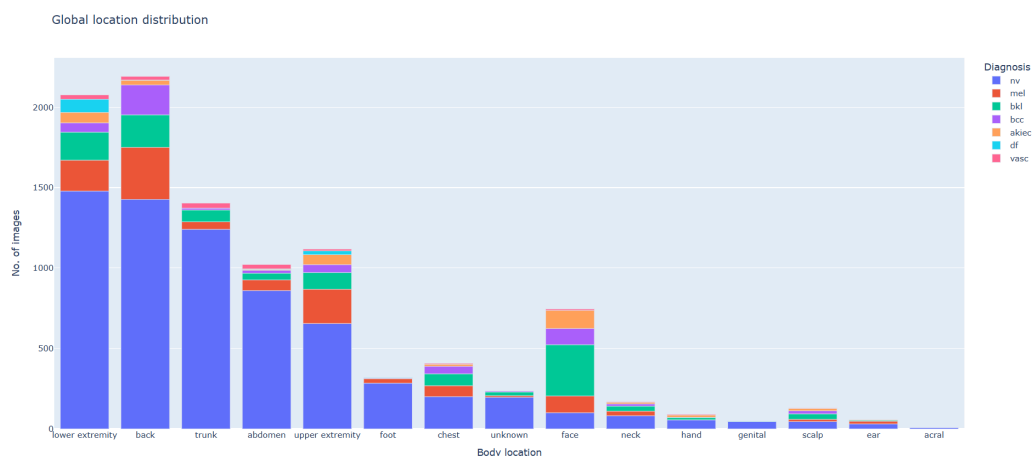
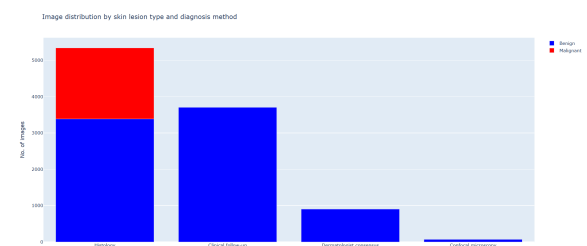
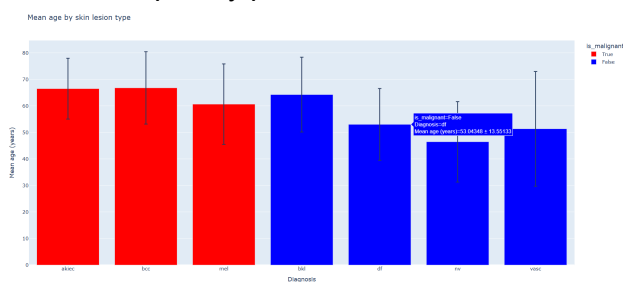
Dataset

We found out that *mel*, *bcc* and *akiec* are malignant skin diseases, while the others are benign. Although the main goal of the project was the 7 class classification, we find it interesting to consider this division in our analysis (between Benign and Malignant), as in medicine Malignant detections tend to be more valuable. As said before the dataset is

highly imbalanced with one main class melanocytic nevi (*nv*) that covers a 67% of the dataset, see on the plot:



Another point we analyzed is the age distribution, where we observed a small tendency of older ages in malign diseases. Then, we observed disease frequency in the diagnosis methods and location. The majority of them were in Histology diagnosis biopsies and located in mid-up body parts.



Preprocessing

We downloaded both zip files and uploaded in the same folder all the images. Then, we created a dataframe from the metadata file and implemented a dictionary relating to each row the corresponding image path. During the initial Exploratory, two corrupted image files were identified and programmatically filtered out to ensure pipeline stability, leaving a final dataset of 10,013 images. All images were resized to 224x224 pixels and normalized using standard ImageNet mean and standard deviation values to enable a fair comparison

between baseline and transfer learning models. To mitigate overfitting and improve model generalization, robust spatial and color data augmentation techniques were applied exclusively to the training set, including random horizontal and vertical flips, random rotations up to 45 degrees, and color jittering. We also created a standard HAM10000Dataset python class for enabling initialization and training processes based on our dataframe.

Data Splits

The dataset was partitioned into Training (80%), Validation (10%), and Test (10%) sets using the stratified configuration to maintain the right proportions and class representation. Because the HAM10000 dataset contains multiple images of the same lesion from different angles, standard random splitting poses a severe risk of data leakage. To resolve this, in the transfer learning phase we introduced a GroupShuffleSplit, grouping by lesion_id. This strict splitting strategy guarantees that all images belonging to a single lesion remain entirely within one specific subset, ensuring that the validation and test metrics reflect true generalization to unseen patients.

2. Models and Optimization

Architectures Implemented

Two primary architectures were implemented to establish a comparative baseline and leverage advanced deep learning techniques:

- Baseline (SimpleCNN): A custom lightweight Convolutional Neural Network consisting of two convolutional blocks (16 and 32 feature maps), each paired with Batch Normalization, ReLU activation, and Max Pooling. An AdaptiveAvgPool2d(1, 1) layer was used to dynamically flatten features, feeding into a fully connected head with Dropout (0.3) (to avoid overfitting) for 7-class classification.
- Transfer Learning (EfficientNet-B2): The core model is an EfficientNet-B2 pre-trained on ImageNet. Its compound scaling provides a highly efficient parameter-to-accuracy ratio suitable for medical textures. The default classifier head was replaced with a custom sequential block to prevent memorization.

Training Procedures and Hyperparameters

The EfficientNet-B2 model was trained using a rigorous two-phase transfer learning strategy:

- Phase 1 (Warm-up): The pre-trained backbone was completely frozen. Only the newly initialized classifier head was trained for 5 epochs using the Adam optimizer with a learning rate of 1e-3. A ReduceLROnPlateau scheduler monitored validation loss to stabilize the initial weight adaptation.
- Phase 2 (Fine-Tuning): The entire network was unfrozen to allow end-to-end training. This phase ran for 15 epochs with a significantly lower learning rate of 1e-4 and a weight decay of 1e-5. A CosineAnnealingLR scheduler gradually reduced the learning rate to 1e-6, allowing the network to make micro-adjustments to the specific dermatological features without destroying the pre-trained ImageNet representations.

Loss Functions and Imbalance Management

The models were optimized using the CrossEntropyLoss function. To address the severe class imbalance, two approaches were tested:

- **Class Weights:** Injecting the inverse class frequencies directly into the loss function to heavily penalize misclassifications on minority classes ($W_c = N / (K * N_c)$).
- **Weighted Random Sampler:** Calculating sampling probabilities per image to construct balanced training batches, ensuring the network evaluates an equal distribution of all 7 classes during each training step.

Interpretability and Deployment

To ensure the model serves as an interpretable clinical tool, Grad-CAM (Gradient-weighted Class Activation Mapping) was integrated to generate heatmaps highlighting the specific visual regions driving the model's predictions. Finally, the best-performing model was wrapped into an API to facilitate seamless external inference on new diagnostic images.

EXPERIMENTS

Block 1: Baseline, SimpleCNN

As a first step, we trained a lightweight convolutional network from scratch to establish a performance floor and validate the data pipeline. This architecture was purposefully designed to be computationally efficient while capturing foundational visual patterns. The *SimpleCNN*, consists of two convolutional blocks (Conv2d - BatchNorm - ReLU - MaxPool) with 16 and 32 filters respectively, followed by a global pooling layer, a dropout layer ($p = 0.3$) to mitigate overfitting, and two fully connected layers (32 - 128 - 7). Each block applies a 3x3 convolution, followed by Batch Normalization to stabilize internal activations and accelerate convergence, a ReLU activation function to introduce non-linearity, and 2x2 Max Pooling to systematically reduce spatial dimensions while providing translation invariance.

We ran the training for 20 epochs with the Adam optimiser ($lr = 1e-3$) and a batch size of 32. A Class Weight strategy was used to address the strong class imbalance in HAM10000 in this approach. One important limitation of this stage is that the train/validation split was performed by image, not by lesion ID. Because the same lesion can appear in multiple images, some data leakage between splits is possible, which may slightly inflate the baseline figures. We corrected this limitation in subsequent experiments.

We expect the SimpleCNN to achieve a modest baseline performance, as its limited architectural capacity (with only 10.000 parameters) will be insufficient to extract the complex visual patterns necessary to combat the severe class imbalance of the clinical dataset.

Block 2: Transfer learning with EfficientNET

In this transfer learning pipeline, we did the following procedures:

- **Shared Phase 1 (head-only warmup).** We loaded EfficientNet-B2 pretrained on ImageNet, froze the entire backbone and replaced the original classifier head with a custom one for our 7 dermatoscopic classes. We trained only the new head for 5 epochs with $lr = 1e-3$, allowing it to adapt to the dataset without disturbing the backbone's pretrained features. The resulting weights were saved which served as the **shared starting point** for the three subsequent experiments. This guarantees that any performance difference between experiments comes purely from the Phase 2 strategy, not from a different warmup state.

- **Experiment 1 — Baseline Phase 2.** Starting from the previous saved model, we unfroze the entire backbone and ran a standard fine-tuning for 15 epochs with $lr = 1e-4$ and $weight_decay=1e-5$, using *CrossEntropyLoss* with class weights to address the dataset imbalance. This experiment establishes the reference performance of full fine-tuning without additional techniques.
- **Experiment 2 — WeightedRandomSampler.** We applied the identical fine-tuning setup as Experiment 1 but replaced the standard DataLoader with a WeightedRandomSampler. This oversamples minority classes melanoma, BCC, AKIEC, dermatofibroma and vascular lesions ensuring each batch contains a more balanced class distribution. The hypothesis was that this would improve recall on the rare malignant classes, which is the most clinically critical metric.
- **Experiment 3 — Adaptive learning rate.** We kept the standard training loop but added a ReduceLROnPlateau scheduler with $patience=3$ and $factor=0.5$. This automatically halves the learning rate whenever validation loss plateaus, allowing the model to settle into finer minima during the later epochs of fine-tuning.

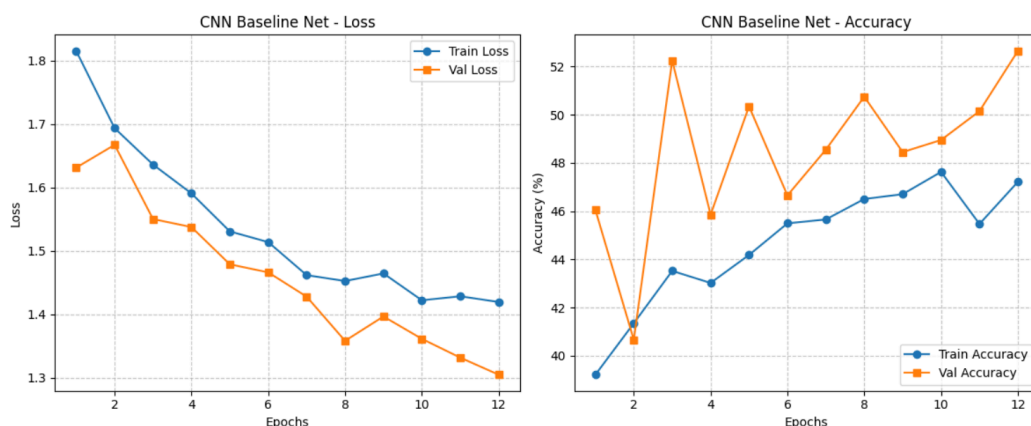
RESULTS

First of all, for our results we considered apart from accuracy some more robust metrics as:

Macro F1-Score: This is our primary metric. It calculates the balance between Precision (not making false alarms) and Recall (finding the actual cases) for each class independently, and then averages them. This forces the model to treat rare diseases (like Melanoma) with the exact same importance as common ones.

In highly imbalanced datasets like HAM10000, relying solely on Accuracy is misleading. A model could achieve a 67% accuracy simply by guessing "Nevus" for every image, which is clinically useless because it would miss 100% of the cancers.

Baseline CNN results:



These curves show how during training, the validation accuracy exhibited extreme volatility, fluctuating erratically across epochs rather than converging smoothly. This instability was directly caused by applying heavy Class Weights to a low-capacity model. The training accuracy marginally improved from 39.21% to 47.22%, while the validation results with a better 52.65% accuracy. This phenomena occurs due to the following reasons:

- Dropout makes the net try to learn on train without all its neurons, on evaluation it disappears.
- Data augmentation modifies images for training which makes it more difficult to learn patterns.
- Class weight implies forcing to learn minor classes in training but not on validation (majority prediction is more beneficial on this simple net).

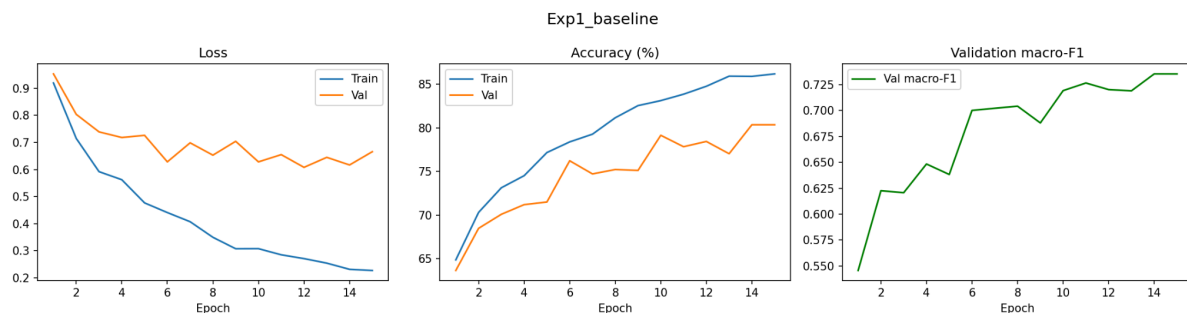
Global Accuracy: 54.09%

	precision	recall	f1-score	support
nv	0.98	0.54	0.70	671
mel	0.29	0.74	0.41	112
bkl	0.35	0.43	0.39	110
bcc	0.33	0.50	0.39	52
akiec	0.13	0.38	0.20	32
vasc	0.50	0.71	0.59	14
df	0.09	0.18	0.12	11
accuracy			0.54	1002
macro avg	0.38	0.50	0.40	1002
weighted avg	0.76	0.54	0.59	1002

Furthermore, the final model achieved a Test Accuracy of 54%, slightly surpassing the best Validation Accuracy of 52%. This 2% difference falls well within standard statistical variance. In general the significantly lower accuracy explicitly highlights the effect of the Class Weights applied to the Cross-Entropy Loss. The 0.4 avg test f1-score shows that the model has not predicted well all classes, specially minor ones as *df* or *vasc* (with a 0.2 and 0.12 f1 respectively) which suggests the low capacity of the model. Also, low Recall for Nevi (0.54) and poor Precision for Melanoma (0.29) are alarmingly low.

Transfer learning

Exp1



Loss curve

The training loss drops consistently from ~0.93 to ~0.22 across the 15 epochs, showing the model is learning effectively. The validation loss also decreases initially but stabilizes around 0.65–0.70 from epoch 5 onwards without following the training curve down. This gap between train and val loss is a sign of mild overfitting. The model is memorizing the training set more than it is generalizing. However the validation loss does not increase sharply, so it remains controlled.

Accuracy curve

Train accuracy climbs steadily from 65% to nearly 85%, while validation accuracy reaches nearly 80% and plateaus from epoch 8 onwards. The 5% gap between the two is consistent with the overfitting observed in the loss curve but remains at an acceptable level.

Validation macro-F1

This is the most important chart. The val F1 starts at 0.55 and grows consistently up to 0.73 by epoch 15, with some noise in the middle epochs particularly the dip around epoch 3–4 which is typical during fine-tuning as the backbone adapts. Crucially, the F1 curve does not plateau or decline at the end, suggesting the model could still improve with more epochs. The val macro-F1 shows a consistent upward trend throughout all 15 epochs without plateauing, suggesting the model kept improving and could have benefited from additional training epoch.

Other metrics:

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=====
Experiment_1_baseline - Test Metrics
=====
Macro F1 : 0.6380
AUC-ROC  : 0.9643

Recall malignant classes:
mel: 0.7034
bcc: 0.8108
akiec: 0.5200

```

	precision	recall	f1-score	support
nv	0.95	0.83	0.89	701
mel	0.47	0.70	0.56	118
bkl	0.66	0.75	0.70	100
bcc	0.65	0.81	0.72	37
akiec	0.52	0.52	0.52	25
vasc	0.47	0.88	0.61	8
df	0.75	0.33	0.46	9
accuracy			0.80	998
macro avg	0.64	0.69	0.64	998
weighted avg	0.83	0.80	0.81	998

Experiment 1 establishes a strong baseline with AUC-ROC of 0.96 and global accuracy of 80%. The main weakness is concentrated in the rare malignant classes particularly mel recall at 0.70 and akiec at 0.52. These numbers directly motivate Experiments 2 and 3, which target precisely those classes through improved sampling and learning rate strategies.

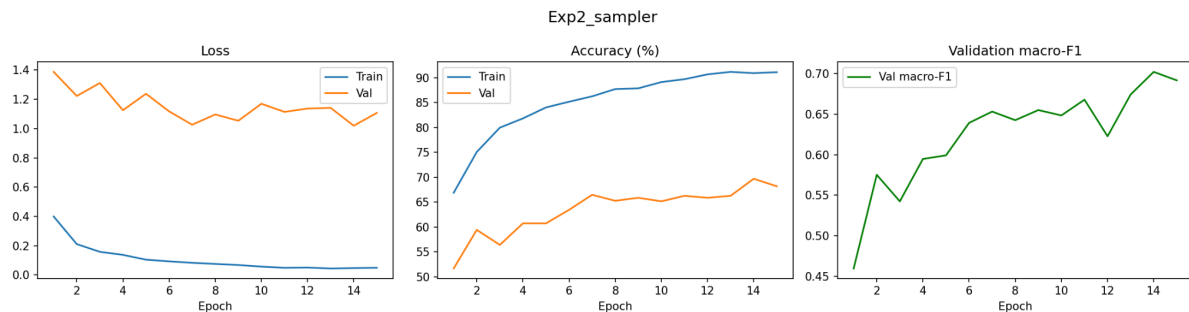
Confusion Matrix:



The confusion matrix confirms two structural problems. First, the mel/nv confusion (21% false negative rate on melanoma) is the most clinically dangerous error and the main

motivation for Experiment 2 with WeightedRandomSampler, which should oversample mel during training. Second, the poor performance on df and akiec is driven by data scarcity rather than model capacity; no architecture would solve this without more training examples.

EXP2



Loss curve

The train loss collapses dramatically from 0.40 to nearly 0.05 much faster and lower than Experiment 1. This is a direct consequence of the WeightedRandomSampler: by oversampling minority classes, each batch is more balanced, so the model finds it easier to minimize the loss on the training set. However the val loss starts at 1.38 and only drops to 1.10, remaining consistently high and noisy throughout. The gap between train (0.05) and val (1.10) is enormous so this is a sign of severe overfitting.

Accuracy curve

Consistent with the loss train accuracy reaches ~91% while val accuracy only reaches 69%. A 22-point gap between train and val is significantly worse than Experiment 1 (~5 points). The val accuracy also starts very low at 51% in epoch 1, indicating the model struggles initially with the real unbalanced distribution of the validation set after being trained on artificially balanced batches.

Validation macro-F1

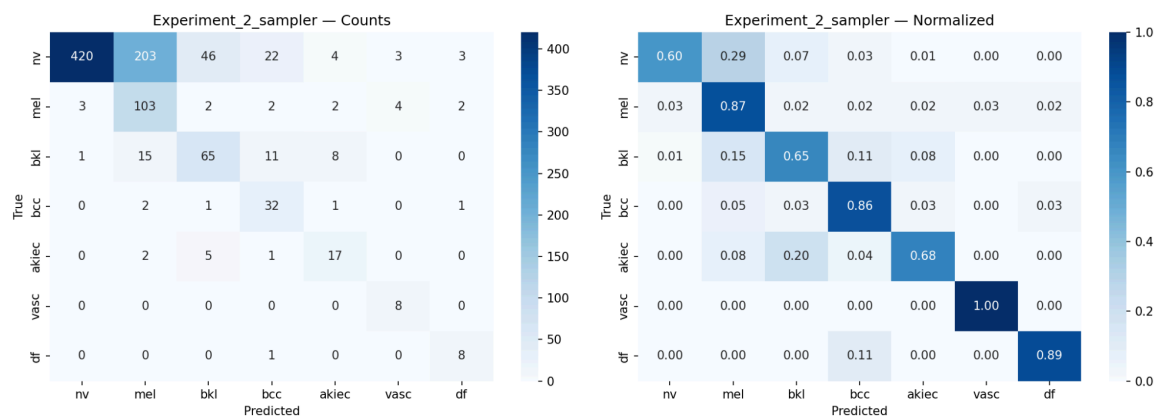
Despite the overfitting, the val F1 reaches 0.70 by epoch 15 with a noisy but upward trend. The noise, particularly the dip around epoch 11–12 reflects the instability caused by the sampler. Due to this instability F1 value is slightly lower in this train-val pipeline than in the previous.

Other metrics:

Experiment_2_sampler — Test Metrics				
Macro F1 : 0.6281				
AUC-ROC : 0.9521				
Recall malignant classes:				
mel: 0.8729				
bcc: 0.8649				
akiec: 0.6800				
	precision	recall	f1-score	support
nv	0.99	0.60	0.75	701
mel	0.32	0.87	0.47	118
bkl	0.55	0.65	0.59	100
bcc	0.46	0.86	0.60	37
akiec	0.53	0.68	0.60	25
vasc	0.53	1.00	0.70	8
df	0.57	0.89	0.70	9
accuracy			0.65	998
macro avg	0.56	0.79	0.63	998
weighted avg	0.83	0.65	0.69	998

Global metrics show a slight regression Macro F1 drops from 0.6380 to 0.6281, AUC-ROC from 0.9643 to 0.9521, and accuracy from 80% to 65%. The WeightedRandomSampler produces a deliberate trade-off: nv recall drops from 0.83 to 0.60, pulling down accuracy and weighted metrics, while every minority class improves significantly. The cost is mel precision falling to 0.32 the model now over-predicts melanoma aggressively, flagging many benign lesions as suspicious.

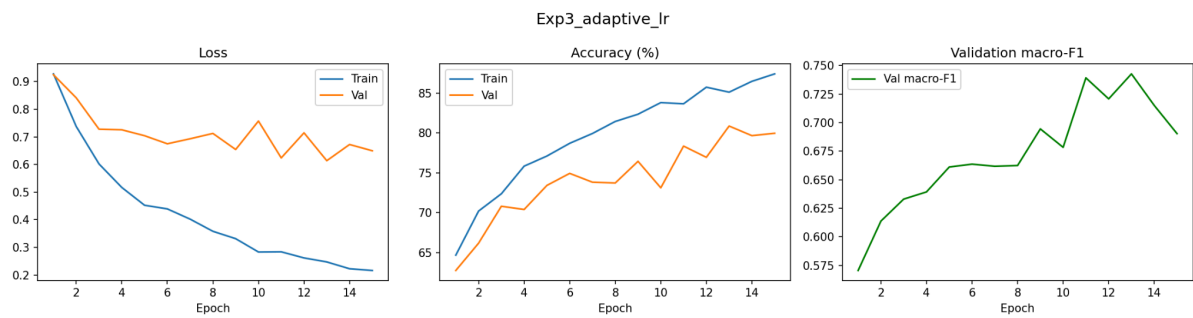
Confusion Matrix:



The most striking finding is the nv row — 203 nevus misclassified as melanoma (29% false positive rate vs 11% in Experiment 1), a direct consequence of the sampler pushing the decision boundary toward minority classes.

On the malignant side the picture is strongly positive. Melanoma false negatives collapse from 25 to just 3 (21% → 3%), the most clinically relevant improvement in the project. BCC holds at 0.86, akiec improves to 0.68 despite persistent confusion with bkl, and df recovers remarkably from 0.33 to 0.89.

EXP3



Loss curve

Train loss drops consistently from ~0.91 to ~0.21 across 15 epochs. Val loss also decreases initially but stabilizes around 0.65 from epoch 5 onwards, creating a significant gap of ~0.44 points. This is clear overfitting the model keeps improving on train while val plateaus. The scheduler prevents the val loss from increasing sharply when it detects plateaus, but it does not close the gap between train and val. The oscillations visible between epochs 8–12 correspond to the scheduler reducing the learning rate in response to those plateaus

Accuracy curve

Train accuracy reaches 91% while val accuracy stabilizes around 80%, a gap of ~11 points. The val accuracy is noisier than Experiment 1, reflecting the learning rate adjustments made by the scheduler in the later epochs. The final val accuracy of 80% is comparable to Experiment 1, suggesting similar generalization on the dominant class despite the overfitting visible in the loss curve

Validation macro-F1

The F1 grows steadily from 0.58 to a peak of ~0.74 around epoch 11–12, then drops back to 0.70 by epoch 15. This peak-then-drop pattern is likely caused by an aggressive lr reduction at that point, pushing the model into a slightly worse minimum for the minority classes

Other metrics:

```

=====
Experiment_3_adaptive_lr - Test Metrics
=====
Macro F1 : 0.6800
AUC-ROC  : 0.9619

Recall malignant classes:
mel: 0.7203
bcc: 0.8108
akiec: 0.4800

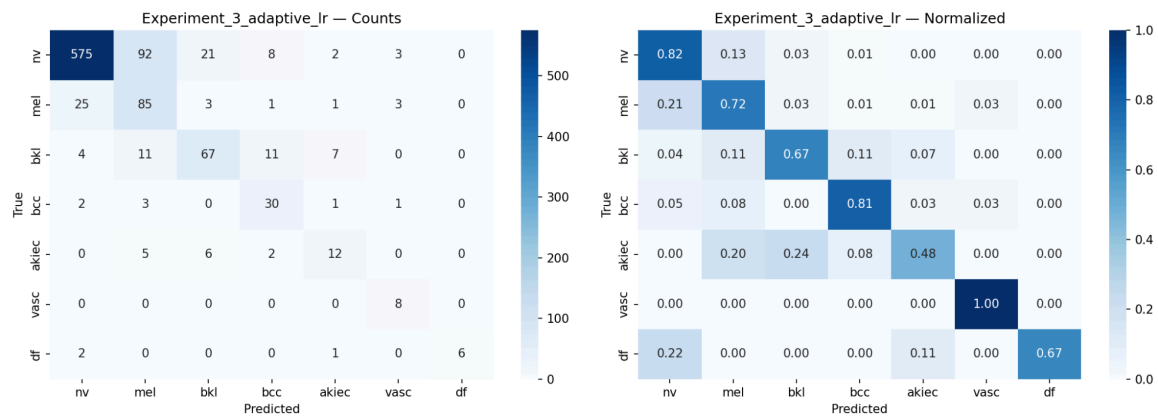
```

	precision	recall	f1-score	support
nv	0.95	0.82	0.88	701
mel	0.43	0.72	0.54	118
bkl	0.69	0.67	0.68	100
bcc	0.58	0.81	0.67	37
akiec	0.50	0.48	0.49	25
vasc	0.53	1.00	0.70	8
df	1.00	0.67	0.80	9
accuracy			0.78	998
macro avg	0.67	0.74	0.68	998
weighted avg	0.83	0.78	0.80	998

Experiment 3 achieves the highest Macro F1 of the three experiments at 0.6800, with AUC-ROC of 0.9619 and accuracy of 78%. Mel recall improves slightly to 0.72 and bcc holds

at 0.81, but akiec drops to 0.48 — the weakest result across all experiments. A notable finding is df reaching precision 1.00, meaning every dermatofibroma prediction was correct, though recall remains at 0.67.

Confusion Matrix:



Experiment 3 closely mirrors Experiment 1 — the adaptive scheduler improves convergence stability but does not change the model's behavior on imbalanced classes. Mel recall improves only marginally to 0.72 with 25 melanomas still missed, akiec drops to 0.48 — the worst result across all three experiments — and df improves to 0.67 but remains far behind Experiment 2. The conclusion is clear: without explicit batch rebalancing, adaptive LR alone is insufficient to address class imbalance.

DISCUSSION

The most significant finding of this project is the clear superiority of Transfer Learning over training from scratch. As hypothesized, the SimpleCNN baseline proved that simply penalizing errors cannot solve the dataset's complexity; the network requires deep representational capacity to truly understand the images. Among our fine-tuning strategies, Experiment 3 (Adaptive Learning Rate) emerged as the most effective model, achieving the highest Macro F1-score (0.68) by dynamically stabilizing the training and mathematically balancing all classes. In contrast, Experiment 2 aggressively combined a WeightedRandomSampler with class_weights, creating an extreme "double penalty" for minority classes. While this caught more melanomas, it triggered severe overfitting and an unacceptable loss of precision, causing the model to misclassify hundreds of healthy nevi as malignant. Due to these clinically unsafe false alarms, Experiment 2 is not a viable solution. Ultimately, Experiment 3 proves that controlled training stability yields much better diagnostic results than excessive mathematical penalization.

CONCLUSIONS

In conclusion, transfer learning is crucial for building AI that can diagnose skin cancer. Although Experiment 3 gave us the best model, simply tweaking settings cannot perfectly fix a dataset that is missing so many examples of rare diseases. To build a truly reliable medical tool in the future, we need to stop relying on code tricks and focus on the data itself—either by collecting naturally balanced datasets or by using advanced AI to generate realistic images of rare cancers.

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